

Intro to Peano Systems

Y'all love Set Theory, don't you?

Mysterious Category Theory Enthusiast

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What is Inductive Set?

Definition 1.

Given a set x , its *Successor* $x^+ := x \cup \{x\}$.

Definition 2.

A set I is an *Inductive Set*, if:

1. $\emptyset \in I$.
2. For any $x \in I$, $x^+ \in I$.

Example 3.

Define $0 := \emptyset$, and $(n + 1) := n^+ = n \cup \{n\}$. This realizes $\mathbb{N}$ as an inductive set.

Properties of Inductive Sets

Proposition 4.

Unions / Intersections of inductive sets is inductive set.

Proof.

Fill in the details on the blackboard :)



Is there a SMALLEST Inductive Set?

Surely there is one, right? Like natural numbers $\mathbb{N}$.

~~**Answer 1:** Yes, take intersection of all inductive sets.~~

~~Yep, this answer is screwed up by ZFC axioms.~~

Answer 1': Kind of: Take S as a set of (arbitrary) inductive sets, define $\omega := \bigcap_{I \in S} I$, this is clearly inductive. Question is, is it the smallest?

ω is the SMALLEST Inductive Set

Fix any inductive set T , define $S_T :=$ all inductive subsets of T . Define ω with respect to $S := S_T$.

Theorem 5.

For each inductive set I , $\omega \subseteq I$.

In particular, $\omega \cong \mathbb{N}$, surprise, surprise...

For any inductive set B , consider $\omega \cap B \subseteq \omega \subseteq T$.

Since $\omega \cap B$ is also inductive, $\omega \cap B \in S_T$, so $\omega \subseteq \omega \cap B$, implying $\omega = \omega \cap B$.

Hence, $\omega \subseteq B$. $\square$

Peano System

Definition 6.

A *Peano System* is a set N , an element $e \in N$, and a function $s : N \rightarrow N$, such that:

- $e \notin s(N)$.
- If $A \subseteq N$, $e \in A$, and $s(A) \subseteq A$, then $A = N$.

Example 7.

The set ω of all finite successors of $\emptyset$, with succeeding function $(-)^+ : \omega \rightarrow \omega$ is a Peano System. In particular, $\omega \cong \mathbb{N}$ as Inductive Sets / Peano Systems.

Moreover, ω is the *initial* Peano System, surprise, surprise...